

Experiment 3

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1 Title

Study of Operational Amplifier (OPAMP) configurations and their arithmetic applications.

2 Aim

The main goal of this experiment is to see how the LM741 operational amplifier works in practice. We will set it up as both inverting and non-inverting amplifiers to check its voltage gain, and then build analog adder and subtractor circuits.

3 Theory

3.1 Operational Amplifier

An operational amplifier (like the LM741 we're using) is basically a high-gain voltage amplifier that takes a differential input. In a perfect world, it has infinite input impedance (so it doesn't draw any current from our signal source) and zero output impedance (so it can drive any load without dropping voltage). We almost always use it with a negative feedback loop to keep the massive open-loop gain stable and controllable.

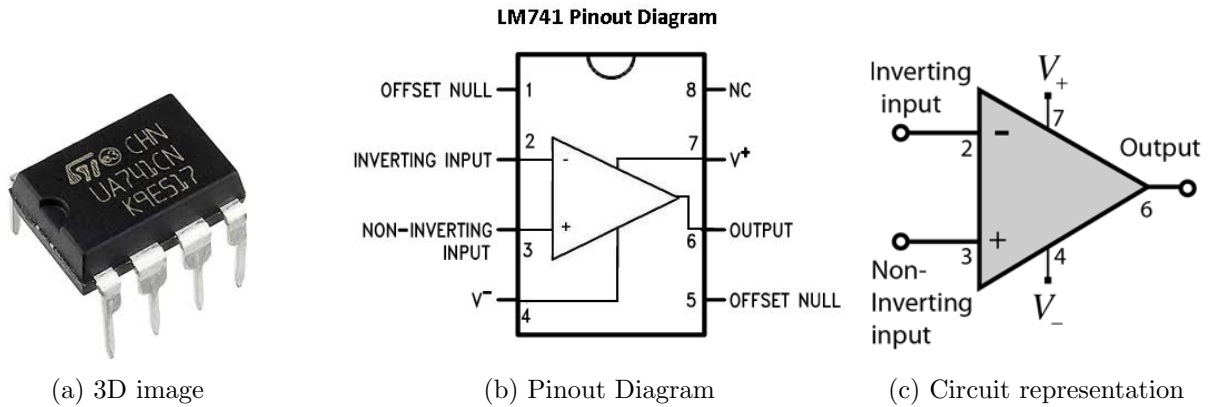


Figure 1: Operational Amplifier LM741

3.2 Inverting Amplifier

For the inverting setup, we ground the non-inverting pin (+) and feed our input signal into the inverting pin (-) through a resistor, R_i . Because of the op-amp's extremely high gain, it forces the voltage difference between its input terminals to zero, creating a "virtual ground" at the inverting pin. Applying Kirchhoff's laws at this node gives us a theoretical gain of:

$$A_{th} = \frac{V_{out}}{V_{in}} = -\frac{R_f}{R_i} \quad (1)$$

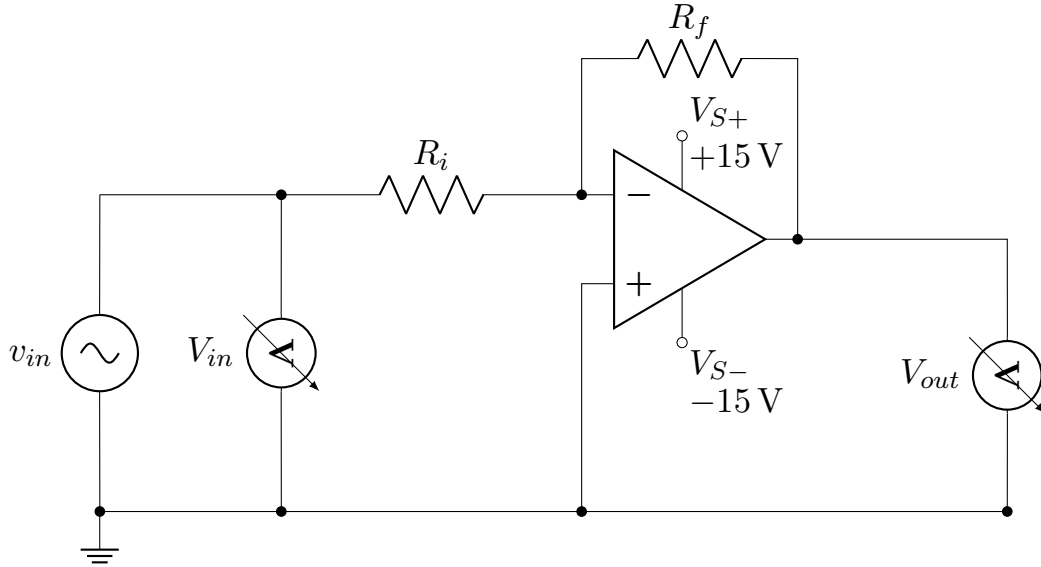


Figure 2: Circuit diagram of an Inverting Amplifier

3.3 Non-inverting Amplifier

Here, we do the opposite and send the input signal directly into the non-inverting terminal. Thanks to the "virtual short" concept, the inverting terminal is forced to match the input voltage. Working out the math, the theoretical gain for this configuration is:

$$A_{th} = \frac{V_{out}}{V_{in}} = 1 + \frac{R_f}{R_i} \quad (2)$$

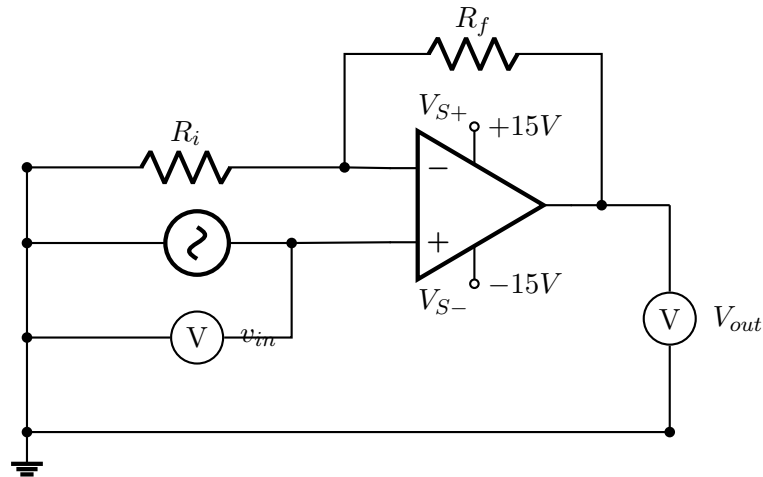


Figure 3: Circuit diagram of a Non-Inverting Amplifier

3.4 Adder (Summing Amplifier)

We can easily modify the inverting amplifier to sum a bunch of voltages. By connecting multiple inputs to the virtual ground node via separate resistors, their individual currents add up. When we make all the resistors equal (in our case, $R_f = R_1 = R_2 = R_3 = 1.0 \text{ k}\Omega$), the output is just the inverted sum of all the inputs:

$$V_{out} = -(V_1 + V_2 + V_3) \quad (3)$$

As we did not have enough different voltage sources, we added 2 resistors (2.15 & 2.2 k Ω) in series with the main source and used their junction points as 3 different sources.

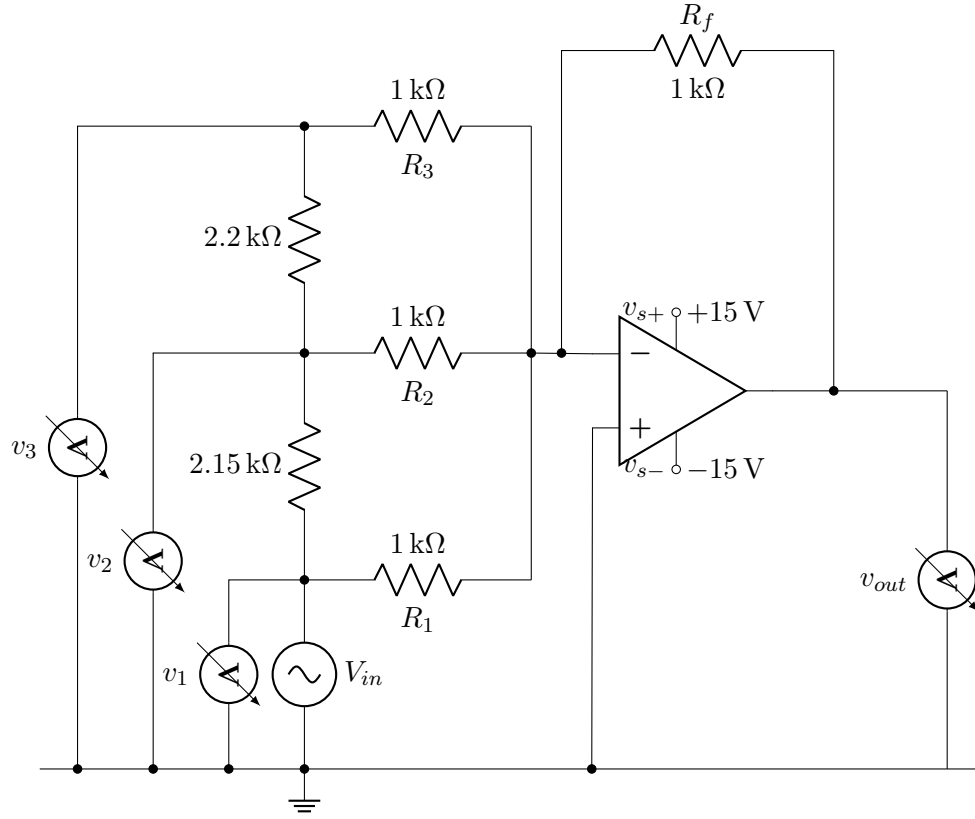


Figure 4: Circuit diagram of OP-Amps as summing Amplifier

3.5 Subtractor (Differential Amplifier)

To find the difference between two signals, we use both input terminals of the op-amp. By balancing the resistor network so that all four resistors are equal ($R_1 = R_2 = R_3 = R_4 = 1.0 \text{ k}\Omega$), the common-mode parts of the signals cancel out. This leaves us with an output that is exactly the difference between the two inputs:

$$V_{out} = V_2 - V_1 \quad (4)$$

Also this time, in order to make 2 different voltage sources, we took a variable resistor and added that with main source, then we changed source voltage as well as the resistor to get different measurements.

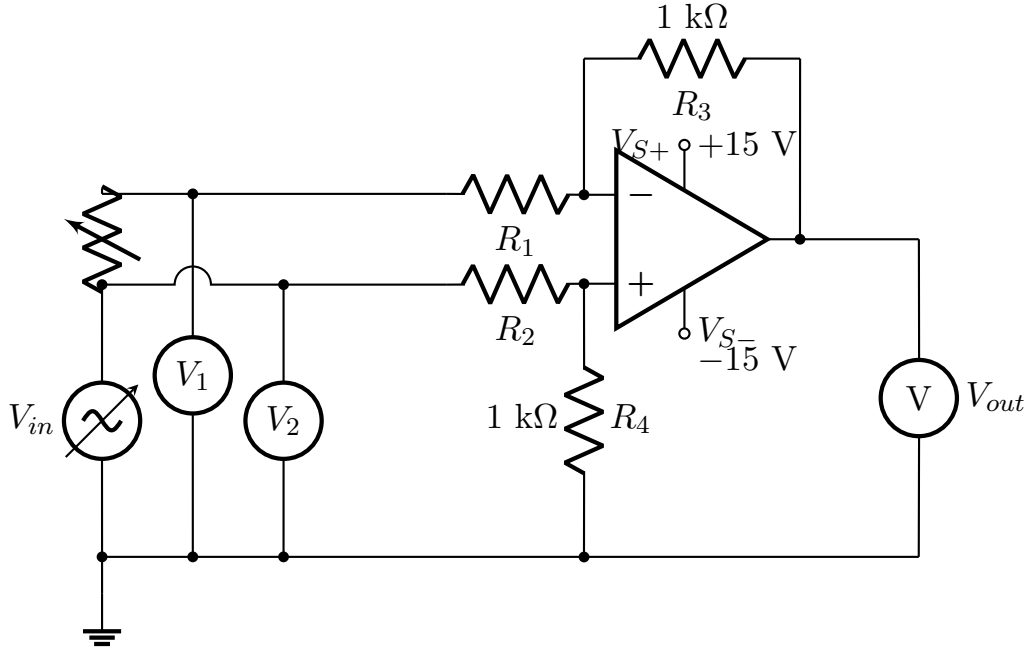


Figure 5: Circuit diagram of OP-Amps as differential Amplifier

4 Data

4.1 Inverting Amplifier Data

Table 1: Experimental Observations for Inverting Amplifier ($R_i = 1.0 \text{ k}\Omega$, $R_f = 9.9 \text{ k}\Omega$)

Sl. No.	V_{in} (V)	V_{out} (V)	Measured Gain (V_{out}/V_{in})	Average Measured Gain	Theoretical Gain ($-R_f/R_i$)
1.	0.2	-2.10	-10.50	-10.04	-9.90
2.	0.5	-4.90	-9.80		
3.	0.7	-7.00	-10.00		
4.	1.0	-10.00	-10.00		
5.	1.1	-10.90	-9.91		
6.	1.2	-12.00	-10.00		

Table 2: Experimental Observations for Inverting Amplifier ($R_i = 1.0 \text{ k}\Omega$, $R_f = 21.8 \text{ k}\Omega$)

Sl. No.	V_{in} (V)	V_{out} (V)	Measured Gain (V_{out}/V_{in})	Average Measured Gain	Theoretical Gain ($-R_f/R_i$)
1.	0.1	-2.10	-21.00	-22.07	-21.80
2.	0.2	-4.50	-22.50		
3.	0.3	-6.80	-22.67		
4.	0.4	-8.90	-22.25		
5.	0.5	-11.00	-22.00		
6.	0.6	-13.20	-22.00		

Table 3: Experimental Observations for Inverting Amplifier ($R_i = 2.2 \text{ k}\Omega$, $R_f = 21.8 \text{ k}\Omega$)

Sl. No.	V_{in} (V)	V_{out} (V)	Measured Gain (V_{out}/V_{in})	Average Measured Gain	Theoretical Gain ($-R_f/R_i$)
1.	0.1	-1.05	-10.50		
2.	0.2	-2.03	-10.15		
3.	0.3	-3.00	-10.00		
4.	0.4	-4.02	-10.05	-10.13	-9.91
5.	0.6	-6.02	-10.03		
6.	0.8	-8.10	-10.13		
7.	1.1	-11.10	-10.09		

Table 4: Experimental Observations for Inverting Amplifier ($R_i = 2.2 \text{ k}\Omega$, $R_f = 9.9 \text{ k}\Omega$)

Sl. No.	V_{in} (V)	V_{out} (V)	Measured Gain (V_{out}/V_{in})	Average Measured Gain	Theoretical Gain ($-R_f/R_i$)
1.	0.1	-0.44	-4.40		
2.	0.5	-2.29	-4.58		
3.	0.9	-4.08	-4.53	-4.52	-4.50
4.	1.3	-5.92	-4.55		
5.	1.7	-7.70	-4.53		
6.	2.1	-9.52	-4.53		

4.2 Non-Inverting Amplifier Data

Table 5: Experimental Observations for Non-Inverting Amplifier ($R_i = 1.0 \text{ k}\Omega$, $R_f = 9.9 \text{ k}\Omega$)

Sl. No.	V_{in} (V)	V_{out} (V)	Measured Gain (V_{out}/V_{in})	Average Measured Gain	Theoretical Gain ($1 + R_f/R_i$)
1.	0.1	1.00	10.00		
2.	0.3	2.89	9.63		
3.	0.5	4.84	9.68	9.69	10.90
4.	0.7	6.74	9.63		
5.	0.9	8.63	9.59		
6.	1.1	10.59	9.63		

Table 6: Experimental Observations for Non-Inverting Amplifier ($R_i = 1.0 \text{ k}\Omega$, $R_f = 21.8 \text{ k}\Omega$)

Sl. No.	V_{in} (V)	V_{out} (V)	Measured Gain (V_{out}/V_{in})	Average Measured Gain	Theoretical Gain ($1 + R_f/R_i$)
1.	0.1	2.32	23.20	21.12	22.80
2.	0.2	4.48	22.40		
3.	0.3	6.06	20.20		
4.	0.4	8.11	20.28		
5.	0.5	10.15	20.30		
6.	0.6	12.23	20.38		

Table 7: Experimental Observations for Non-Inverting Amplifier ($R_i = 2.2 \text{ k}\Omega$, $R_f = 21.8 \text{ k}\Omega$)

Sl. No.	V_{in} (V)	V_{out} (V)	Measured Gain (V_{out}/V_{in})	Average Measured Gain	Theoretical Gain ($1 + R_f/R_i$)
1.	0.1	1.17	11.70	11.22	10.91
2.	0.3	3.38	11.27		
3.	0.4	4.44	11.10		
4.	0.6	6.61	11.02		
5.	0.8	8.88	11.10		
6.	1.0	11.12	11.12		

Table 8: Experimental Observations for Non-Inverting Amplifier ($R_i = 2.2 \text{ k}\Omega$, $R_f = 9.9 \text{ k}\Omega$)

Sl. No.	V_{in} (V)	V_{out} (V)	Measured Gain (V_{out}/V_{in})	Average Measured Gain	Theoretical Gain ($1 + R_f/R_i$)
1.	0.1	0.61	6.10	5.65	5.50
2.	0.5	2.81	5.62		
3.	0.9	5.03	5.59		
4.	1.3	7.21	5.55		
5.	1.7	9.38	5.52		
6.	2.0	11.09	5.55		

4.3 Adder Circuit Data

Table 9: Experimental Observations for Adder Circuit ($R_1 = R_2 = R_3 = R_f = 1k\Omega$)

Sl. No.	V_1 (V)	V_2 (V)	V_3 (V)	Measured V_{out} (V)	Theoretical V_{out} (V) $-(V_1 + V_2 + V_3)$
1.	1.56	0.41	0.13	-2.03	-2.10
2.	2.74	0.72	0.24	-3.58	-3.70
3.	4.21	1.11	0.37	-5.50	-5.69
4.	5.73	1.51	0.51	-7.47	-7.75
5.	7.27	1.91	0.64	-9.46	-9.82
6.	8.63	2.27	0.77	-11.24	-11.67
7.	5.03	1.33	0.45	-6.57	-6.81
8.	2.25	0.60	0.19	-2.95	-3.04
9.	3.55	0.99	0.32	-4.63	-4.86
10.	6.42	1.70	0.58	-8.38	-8.70

4.4 Subtractor Circuit Data

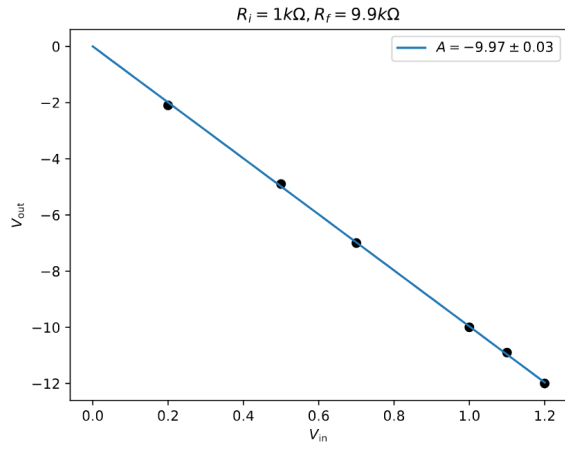
Table 10: Experimental Observations for Subtractor ($R_1 = R_2 = R_3 = R_f = 1k\Omega$)

Sl. No.	V_1 (V)	V_2 (V)	Measured V_{out} (V)	Theoretical V_{out} (V) $(V_2 - V_1)$
1.	3.93	5.22	1.38	1.29
2.	4.08	8.07	4.08	3.99
3.	6.60	8.22	1.75	1.62
4.	3.31	6.56	3.32	3.25
5.	5.75	11.37	5.75	5.62
6.	8.50	11.31	2.99	2.81
7.	2.83	3.77	1.00	0.94
8.	14.54	16.79	2.54	2.25
9.	12.62	16.78	4.43	4.16
10.	5.71	7.58	2.00	1.87
11.	4.88	6.46	1.70	1.58

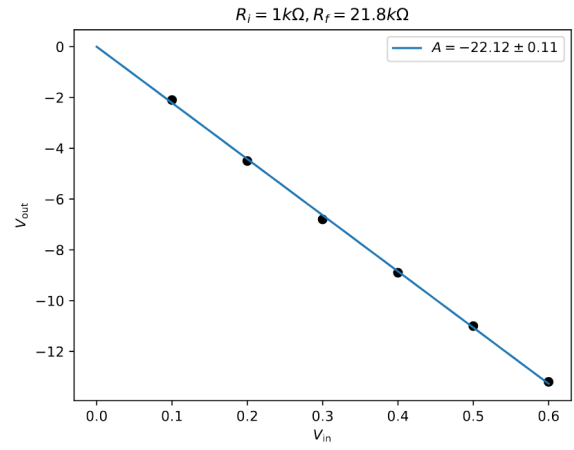
5 Analysis

5.1 Analyzing Inverting and Non-Inverting Amplifiers

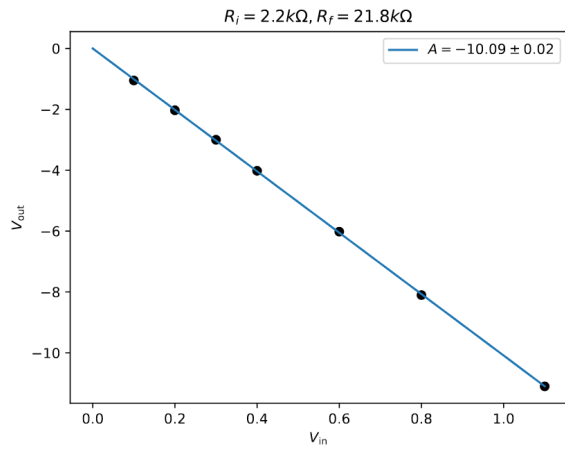
We calculated the experimental gain ($A_{exp} = V_{out}/V_{in}$) for different input voltages across four separate resistor combinations. Taking the average of these measured gains, we compared them directly to our theoretical expectations. The experimental data shows a very strong, consistent correlation with the theory, confirming that the LM741 performs reliably in these standard closed-loop configurations.



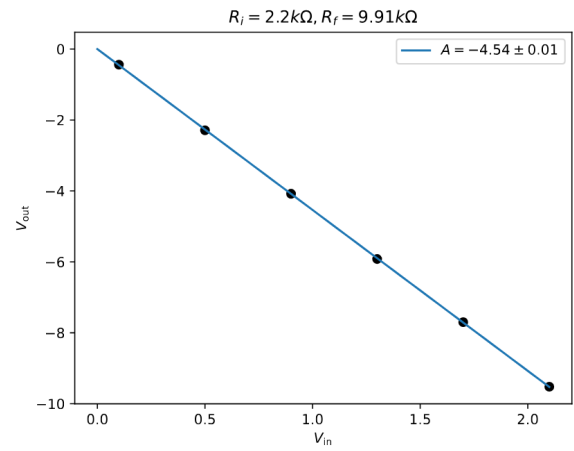
(a) $R_i = 1.0\text{ k}\Omega, R_f = 9.9\text{ k}\Omega$



(b) $R_i = 1.0\text{ k}\Omega, R_f = 21.8\text{ k}\Omega$

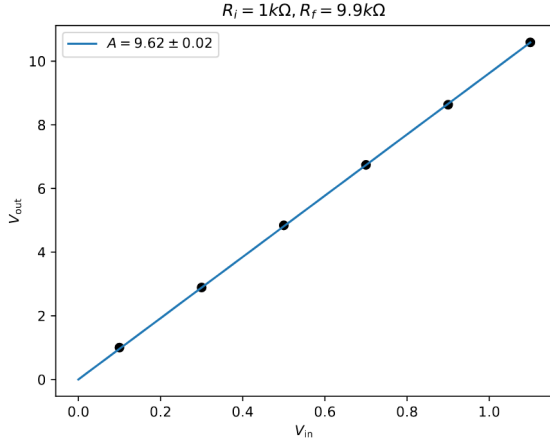


(c) $R_i = 2.2\text{ k}\Omega, R_f = 21.8\text{ k}\Omega$

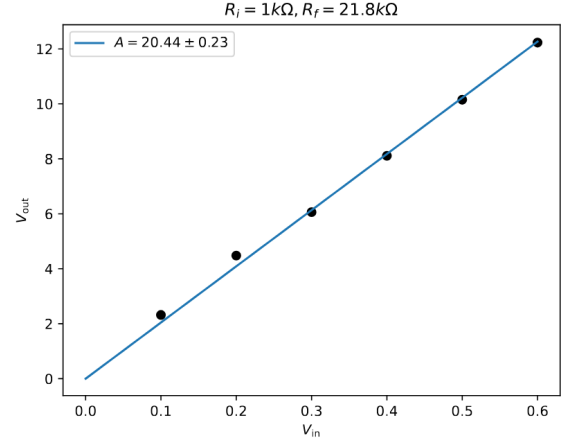


(d) $R_i = 2.2\text{ k}\Omega, R_f = 9.9\text{ k}\Omega$

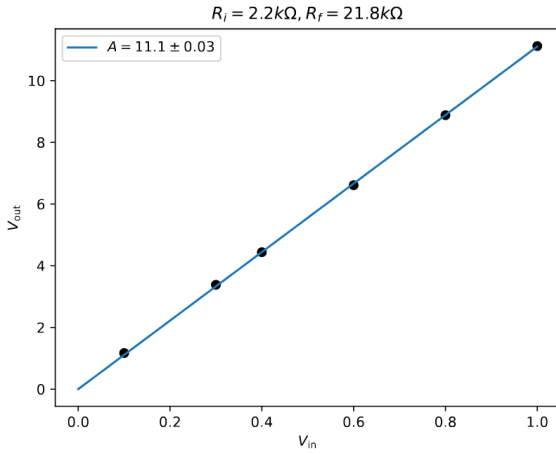
Figure 6: Input vs output voltage plots for the Inverting Amplifier



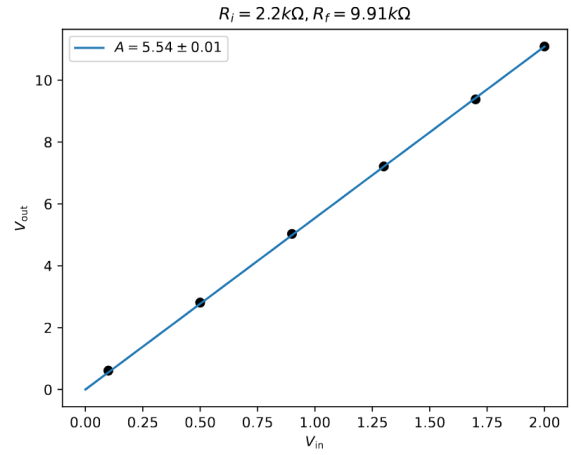
(a) $R_i = 1.0\text{ k}\Omega$, $R_f = 9.9\text{ k}\Omega$



(b) $R_i = 1.0\text{ k}\Omega$, $R_f = 21.8\text{ k}\Omega$



(c) $R_i = 2.2\text{ k}\Omega$, $R_f = 21.8\text{ k}\Omega$



(d) $R_i = 2.2\text{ k}\Omega$, $R_f = 9.9\text{ k}\Omega$

Figure 7: Input vs output voltage plots for the Non-Inverting Amplifier

5.2 Analyzing Adder and Subtractor Circuits

By systematically tuning the input voltages on the breadboard, we logged the observed output voltage (V_{obs}) and compared it against the theoretical calculation (V_{th}). The adder successfully summed and inverted the three distinct inputs, while the subtractor tracked the voltage difference nicely across a wide range of values.

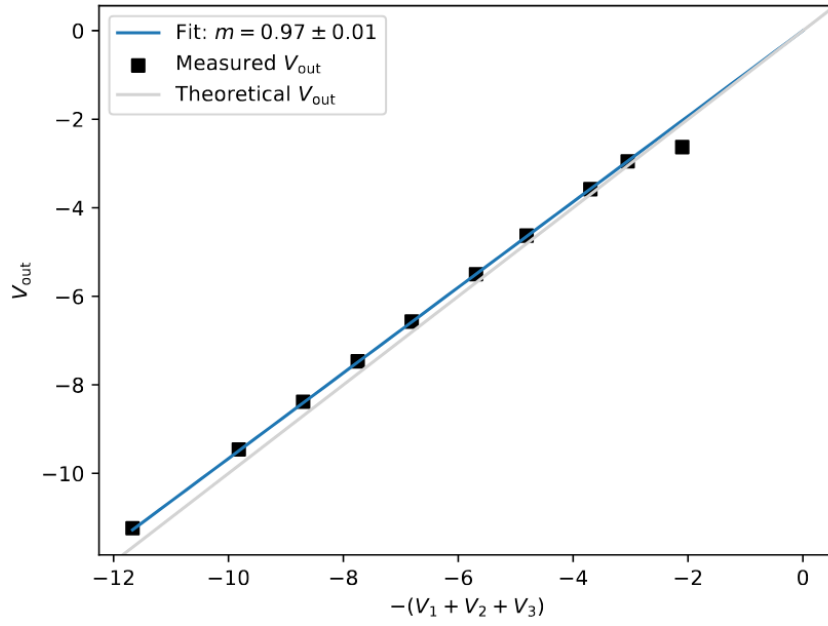


Figure 8: Comparison between V_{obs} and V_{th} in the Adder circuit

Margin of error for adder = $\frac{|1-0.97|}{1} = 3\%$

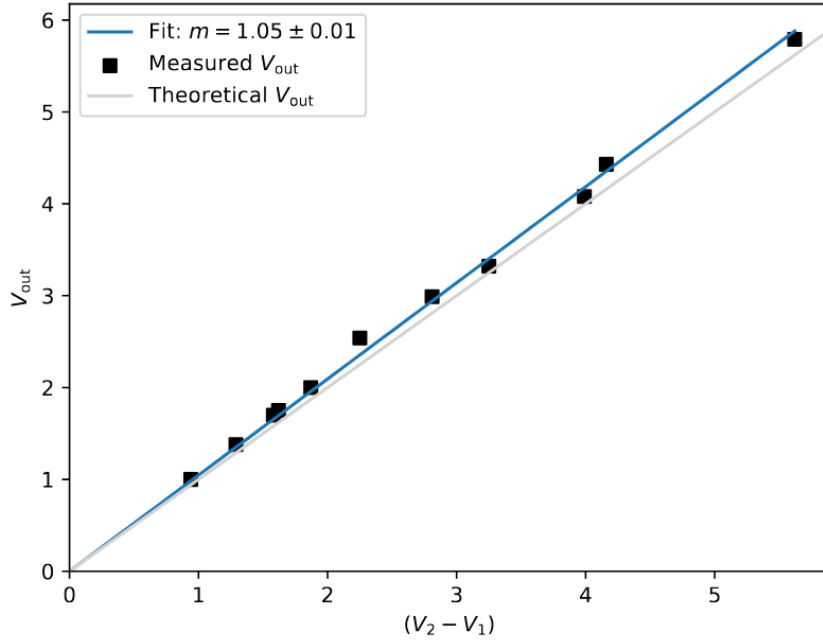


Figure 9: Comparison between V_{obs} and V_{th} in the Subtractor circuit

Margin of error for subtractor = $\frac{|1-1.05|}{1} = 5\%$

6 Sources of Error

- Loose connections or improper placement of wires on the breadboard could introduce additional resistance and voltage drops.

- Fluctuations in the output of the given voltage source.
- Non-ideal behavior of the LM741 OP-AMP (input offset voltage, bias current, finite input impedance).
- Assumption of exactly equal resistances in the adder/subtractor circuits, which deviate slightly in reality.

7 Conclusion

This lab successfully demonstrated the versatility of the LM741 operational amplifier. We verified that it functions predictably as both an inverting and non-inverting amplifier. We also confirmed its utility in analog computation; the adder circuit accurately provided the inverted sum of three signals, and the differential setup successfully isolated the difference between two inputs. The small percentage errors we encountered were expected.